

CMSE 801 - Day 5



Getting more comfortable with writing functions
through practice

September 11, 2024

Heads up: I've been thinking about trying out assigned groups (to make attendance tracking easier). I'll be sending out a poll to get your thoughts about this.

General Course Announcements

- Reminder: Homework 1 is available on the course website. This assignment will be **due at 11:59pm on Friday, September 27.**
 - This is an opportunity to showcase all of the Python fundamentals you've picked up in course and to practice finding useful examples and resources on the web.
 - **The assignment might end up feeling a little tedious (sorry!) but it should be useful and nothing too challenging should come up.** *Give yourself time to work through and consider doing it in multiple sittings.*
 - There is a D2L submission folder for when you complete the assignment

General Course Announcements (cont'd)

- I will be inviting you to a new AI tool that I'm piloting to support your efforts in this course, **HiTA**.
 - HiTA employs a technology called **Retrieval-Augmented Generation (RAG)** to provide context to a **Large Language Model (LLM)** based on course materials, instructor's instructions, and conversation history
 - Details on how to access and use the tool will be provided as well.
- Reminder that office hours and locations are on the website:
<https://cmse.msu.edu/cmse801>
- Remember: at the end of class **you will be uploading your in-class notebook to D2L**.

Questions/comments from Day 5 PCA

- How does the `return` statement work again? When/Why would I use it?
- Some questions about setting default values in functions
- How do I pass lists of values as inputs?
- Questions about the last question (Part 5) -- **we'll look at that at the end**

Let's look at some of the things we can do with functions by taking the simple "Happy Birthday" example

```
In [1]: # Define a function that prints "Happy Birthday to <person>" where the  
def happy_birthday(name):  
    print("Happy birthday to", name)  
happy_birthday("Ian")
```

Happy birthday to Ian

```
In [5]: # What if I want there to be a default name?  
def happy_birthday(name='you!'):  
    print("Happy birthday to", name)  
happy_birthday()
```

Happy birthday to you!

```
In [9]: # What if I want to return the statement instead of just print it?  
def happy_birthday(name='you!'):  
    return("Happy birthday to"+" "+name)  
wishes = happy_birthday("Sparty")  
print(wishes)  
#happy_birthday("Ian")
```

Happy birthday to Sparty

Let's work through Part 5...

Do the following:

1. Write function called `x_squared`.
2. The function should take in a list of x -values of any length, but should have a default of `[1,2,3]`.
3. The function should **return** a new list of values that are x^2 values of the original input values.
4. Print the resulting new list.

```
In [10]: def x_squared(x_vals=[1,2,3]):  
          x2 = []  
          for x in x_vals:  
              x2.append(x**2)  
          return x2
```

```
In [11]: # Let's test it out  
xs = [2,4,6,8,10]  
  
squared = x_squared(xs)  
print(squared)  
  
# Let's also confirm that my default ([1,2,3]) is working:  
default_squared = x_squared()  
print (default_squared)  
  
[4, 16, 36, 64, 100]  
[1, 4, 9]
```


What if we wanted to use a different function *inside* this function?

For example, I could use my `happy_birthday` function from before (though it might not be the obvious choice...)

```
In [12]: def x_squared(vals=[1,2,3]):  
          x2 = [] # Initialize the list to return  
          for x in vals: # Looping by ???  
              x2.append(x**2)  
              if x**2 > 50:  
                  print(happy_birthday("someone!"))  
          return x2  
  
squared = x_squared([2,4,6,8,10])  
print(squared)
```

```
Happy birthday to someone!  
Happy birthday to someone!  
[4, 16, 36, 64, 100]
```

```
In [13]: # I could also pass a default value for when I call the happy_birthday :  
def x_squared(vals=[1,2,3], hb='someone else!'):  
    x2 = [] # Initialize the list to return  
    for x in vals: # Looping by ???  
        x2.append(x**2)  
        if x**2 > 50:  
            print(happy_birthday(hb))  
    return x2  
  
squared = x_squared([2,4,6,8,10])  
print(squared)
```

```
Happy birthday to someone else!  
Happy birthday to someone else!  
[4, 16, 36, 64, 100]
```

Goals for today

- Practice writing functions to structure and compartmentalize your code and make it **reusable**.
 - This will sometimes involve using functions *inside* other functions
- Experiment with how you can use function to start building a "rideshare app" (not really, but go with it..)

Looking forward - Day 6 PCA: Using modules in Python (specifically numpy and matplotlib)

- Explore how you can "import" modules into your notebooks to gain access to additional packages and new functionality
 - `matplotlib` -- finally we can start visualizing values and data! this is the most common library for making plots in Python
 - `numpy` -- you'll primarily start using it to make it easier to do math in Python, but we'll also use it to make working with data easier

